



Dynamics of algorithmic content amplification on TikTok

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Abstract

Content exposure online is increasingly shaped by socio-technical systems, where human and algorithmic factors interact. TikTok exemplifies strong algorithm-driven curation, tailoring the stream of content towards users' interests based on their behavior on the platform. Despite growing concerns about TikTok's societal impact, the dynamics of content amplification at the individual level remain largely unquantified. How quickly, and to what extent, does TikTok amplify content aligned with users' interests? To address these questions, we conduct a sock-puppet audit, deploying bots with different interests to engage with TikTok's "For You" feed. Our findings reveal that content aligned with the bots' interests undergoes strong amplification, with rapid reinforcement typically occurring within the first 200 videos watched. While amplification is consistently observed across all interests, its intensity and dynamics vary, as revealed by time series analyses and Markov models. Although TikTok's algorithm preserves some content diversity, we find a strong negative correlation between amplification and exploration: as the amplification of interest-aligned content increases, the diversity of encountered hashtags declines sharply. Our findings contribute to discussions on socio-algorithmic feedback loops in the digital age and the trade-offs between personalization and content diversity.

Keywords: Social media; TikTok; Algorithmic amplification; Exploration; Sock-puppet audit

1 Introduction

The information we encounter online is increasingly shaped by algorithmic systems that arise from the interplay between human behavior and algorithmic processes [1, 2]. A prominent example is social media platforms, where users are exposed both to content from their social connections, i.e., neighbors in an underlying social network, and to algorithmically mediated content, which originates outside their immediate social circles but is tailored to align with their inferred interests. While the degree to which algorithmic curation influences information flows on platforms varies [3], algorithmic systems were shown to profoundly affect cognitive processes at the individual level, including emotional states and decision-making [4], but also shape the visibility [5], dissemination [6], and consumption of online content [7] on a system-level. As such, algorithmic systems may also affect the emergence and evolution of cultural artifacts [8, 9], making the study of algo-

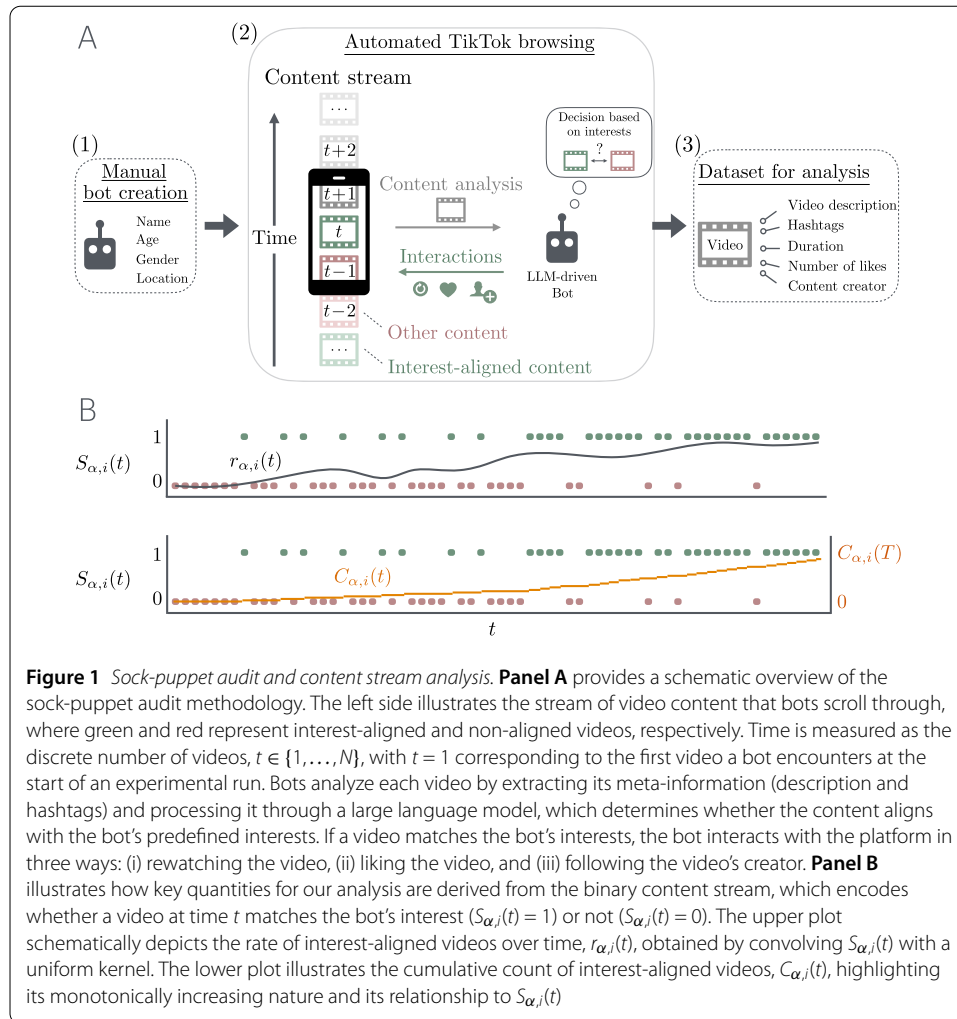
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rhythmically mediated information a crucial frontier in understanding society's transition into the digital age [10, 11].

TikTok represents an extreme case of an algorithm-driven ecosystem [12, 13]. As a short-form video-sharing platform, TikTok allows users to create, share, and engage with highly viral, user-generated content across a wide range of topics, including entertainment, lifestyle, education, and politics. Its “For You” feed curates each user's content diet primarily based on their explicit and implicit interactions with previously encountered videos, such as rewatching, liking, or following content creators. This direct feedback loop enables the algorithm to rapidly and continuously adapt to user preferences and emerging collective trends. Social media platforms typically operate along a continuum between human- and algorithm-driven content curation. Other platforms (including Youtube and Snapchat) have also shifted further toward algorithmic curation, where content exposure is increasingly determined by an algorithmic feed that infers user interests and proactively tailors content accordingly. This growing reliance on algorithmic personalization echoes long-standing concerns about the emergence of “filter bubbles,” in which individuals become intellectually isolated due to highly customized information streams and fully automated recommendation systems [14].

With this in mind, TikTok's rapid global expansion, combined with its highly personalized user experience, raises critical questions about its societal impact. Previous studies have examined various aspects of TikTok's algorithmic feed, including its potentially addictive nature [15–17], its influence on social connectedness [18], and the role of user-algorithm feedback loops in shaping content exposure [19–21]. Despite these insights, it remains largely unclear how—and to what extent—TikTok's recommendation algorithm shapes the content distribution individuals encounter by amplifying content aligned with their interests.

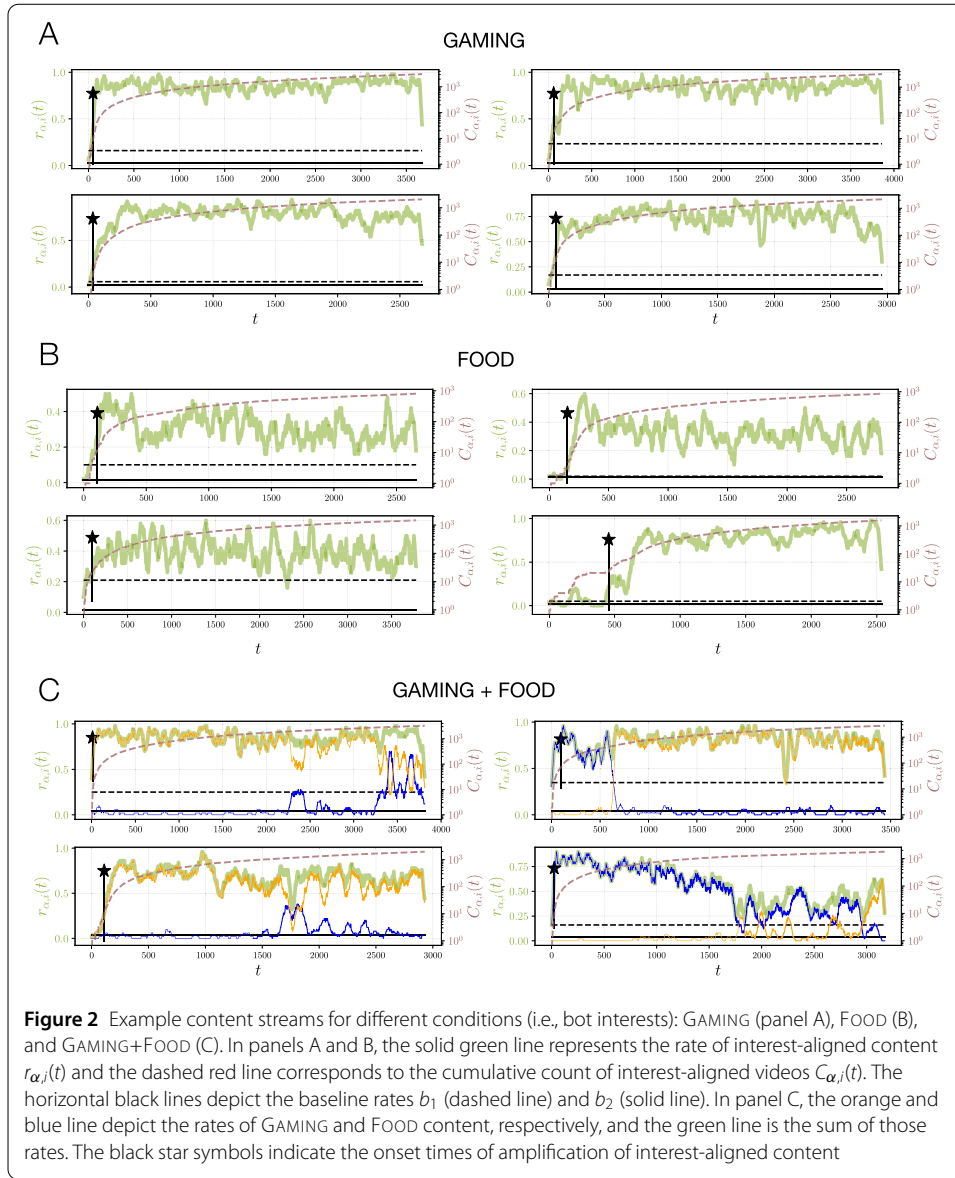
To address this gap, we conduct a sock-puppet audit by deploying automated bots that interact in real-time with TikTok's “For You” feed, each programmed with distinct content preferences (or interests). Extending beyond previous sock-puppet methodologies [2, 22–24], our approach leverages a large language model (GPT-3.5 Turbo) to dynamically assess the relevance of a video based on its description and its associated hashtags. Specifically, bots evaluated each video encountered to determine whether it matched their assigned interests. Upon detecting alignment, the bots performed predefined user-like interactions, i.e., rewatching the video, liking it, and following the account of the video's creator. Investigating content amplification on TikTok requires selecting topic categories that are sufficiently prevalent to ensure that the bots can effectively signal their interests within a reasonable timeframe. To this end, we analyze three experimental conditions: (i) bots with a single interest in GAMING content (condition G), (ii) bots with a single interest in FOOD content (F), and (iii) dual-interest bots engaging with both GAMING and FOOD content ($G + F$). The dual-interest bots interact with content classified as either GAMING, FOOD, or both. We selected GAMING and FOOD because they are both (i) highly prevalent on TikTok, which ensures that our bots encounter a sufficient number of aligned content within a reasonable time frame, and (ii) relatively neutral domains, in contrast to political content, where platform behavior and user engagement introduce additional biases that we aimed to avoid. The combined-interest (GAMING+FOOD) condition further allows us to investigate how the platform responds when users express multiple interests. See Fig. 1



for a schematic overview of our sock-puppet methodology and the Methods section for details on its implementation.

This sock-puppet methodology gives us the opportunity to investigate how TikTok’s feed algorithm shapes the content distribution based on the pre-specified behavior of bots. Specifically, we investigate how quickly the recommendation algorithm responds a bot’s signaled interests and how it adapts over time. Additionally, we analyze how the dynamics of content amplification vary across different bot interests and how amplification correlates with other content characteristics, such as the popularity and the duration of videos. Finally, we explore the relationship between content amplification and content diversity, assessing whether increased reinforcement of interest-aligned content narrows down the scope of content exploration.

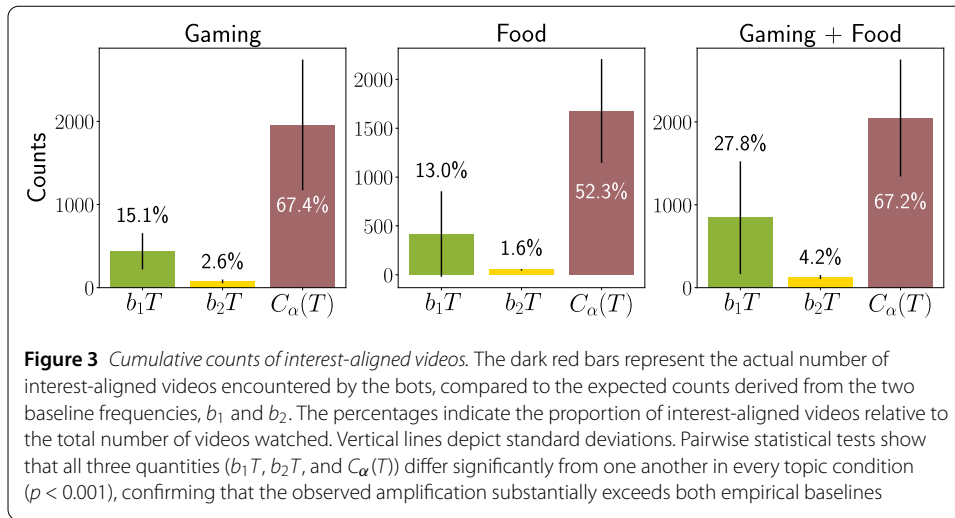
The remainder of the paper is organized as follows. In the Sect. 2, we present our results, detailing the dynamics of interest-aligned content amplification on TikTok’s “For You” feed. We characterize these dynamics using time series analysis and Markov models. Following this, we discuss our findings in the context of existing research on algorithmic feeds and outline the limitations of our study. A detailed description of our sock-puppet methodology, the behavior of the bots, and the computational analysis can be found in Sect. 4 (Methods) and in the Supplementary Material (SM).



2 Results

Content amplification The data from each experiment is represented as a binary discrete time series, $S_{\alpha,i}(t) \in \{0, 1\}$, where $t \in \{1, \dots, T\}$ denotes the index of a video encountered by a bot, from the first video ($t = 1$) to the last ($t = T$). The letter α specifies the experimental condition ($\alpha \in \{G, F, G+F\}$), and i identifies the experimental realization ($i \in \{1, \dots, 14\}$). For example, $S_{G,3}(t)$ represents the time series of the third experimental realization in the GAMING condition. If the 5th video in this experiment matched the bot's interest (i.e., GAMING), then $S_{G,3}(5) = 1$; otherwise, $S_{G,3}(5) = 0$. The frequency of interest-matching (or *aligned*) videos over time, $r_{\alpha,i}(t)$, is calculated as the time-moving average of $S_{\alpha,i}(t)$, i.e., the discrete convolution $r_{\alpha,i}(t) = \sum_{k=-\lfloor w/2 \rfloor}^{\lfloor w/2 \rfloor} \frac{1}{w} S_{\alpha,i}(t-k)$, where w is the window size of the averaging kernel.

In Fig. 2, we show the results of a subset of our 42 experiments (14 runs $\times$ 3 conditions). Panels A and B correspond to the two single-interest conditions GAMING and FOOD, re-

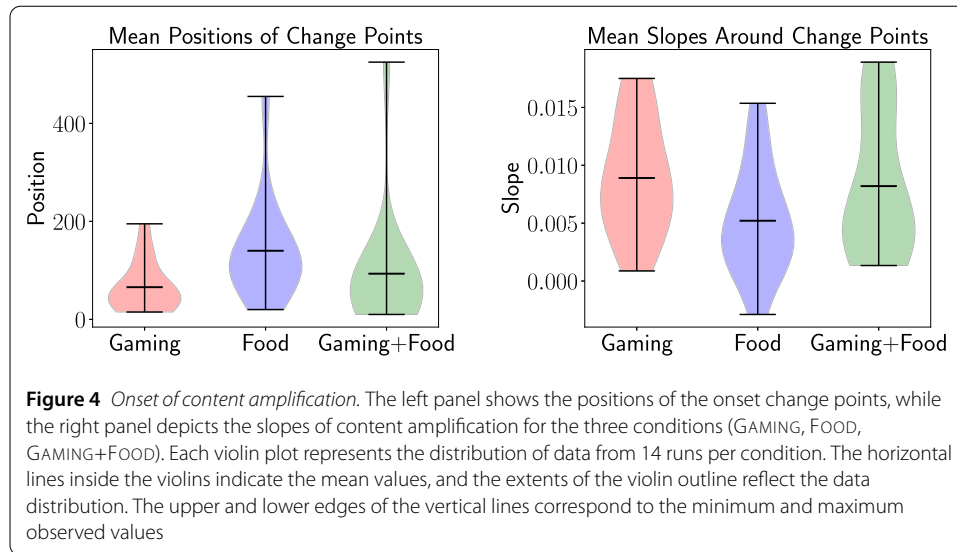


spectively, and panel C shows results of the dual-interest condition GAMING+FOOD. The rate of aligned content, $r_{\alpha,i}(t)$, is depicted as solid green line and the dashed red line represents the cumulative sum of aligned videos encountered by the bot up to time t , defined as $C_{\alpha,i}(t) = \sum_{t=1}^T S_{\alpha,i}(t)$. The total number of aligned videos encountered by a bot in a given experimental run is denoted as $C_{\alpha,i}(T)$; the quantity $C_{\alpha,i}(t)$ is displayed on a logarithmic scale on the right y-axis of each plot in Fig. 2. In panel C, for the dual-interest condition (GAMING+FOOD) we depict three rates in different colors: the rate of GAMING content (orange), FOOD content (blue), and the sum of these two rates (green).

The horizontal black lines serve as baselines for the rate of aligned content. First, the black dashed line represents the baseline b_1 , which corresponds to the average rate of aligned content during the initial phase of the experiment, up to the onset point t_o (black star symbols), before a significant amplification of aligned content. For each experimental condition α and run i , this baseline is computed as $b_1^{\alpha,i} = C_{\alpha,i}(t_o)/t_o$. Second, the black solid line represents the baseline b_2 , which corresponds to the average rate of the content of topic α appearing in experiments where bots are not interested in the topic α . In particular, b_2 for the GAMING condition is defined as the average rate of GAMING content in the FOOD condition, i.e., $b_2^G = \langle C_{F,j}(T) \rangle_j$ (averaged over all experimental realizations j), and vice versa. The baseline frequency b_2^α does not vary across experimental realizations. For the dual-interest condition (GAMING+FOOD) the baseline b_2 is computed as the mean of the single-interest baselines, b_2^α , i.e., $b_2^{G+F} = (b_2^G + b_2^F)/2$. Further details on the baselines are provided in Sect. 4.

We observe a pronounced and rapid amplification of content aligned with the bots' interests for both single-interest bots (panels A and B) and dual-interest bots (panel C). This amplification is often reflected in the sudden and steep increase of $r_{\alpha,i}(t)$ shortly after the start of an experiment, followed by its sustained elevation at later times compared to both baselines ($b_1^{\alpha,i}$ and b_2^α).

The strong amplification of interest-aligned content is further evident when comparing the average cumulative count of interest-aligned videos in each condition, defined as $C_\alpha(T) = \langle C_{\alpha,i}(T) \rangle_i$, to the expected counts based on the baselines b_1 and b_2 , as shown in Fig. 3. Across conditions, the total number of interest-aligned videos consistently surpasses the baseline expectations b_1T and b_2T . For single-interest bots, the amplification is most pro-



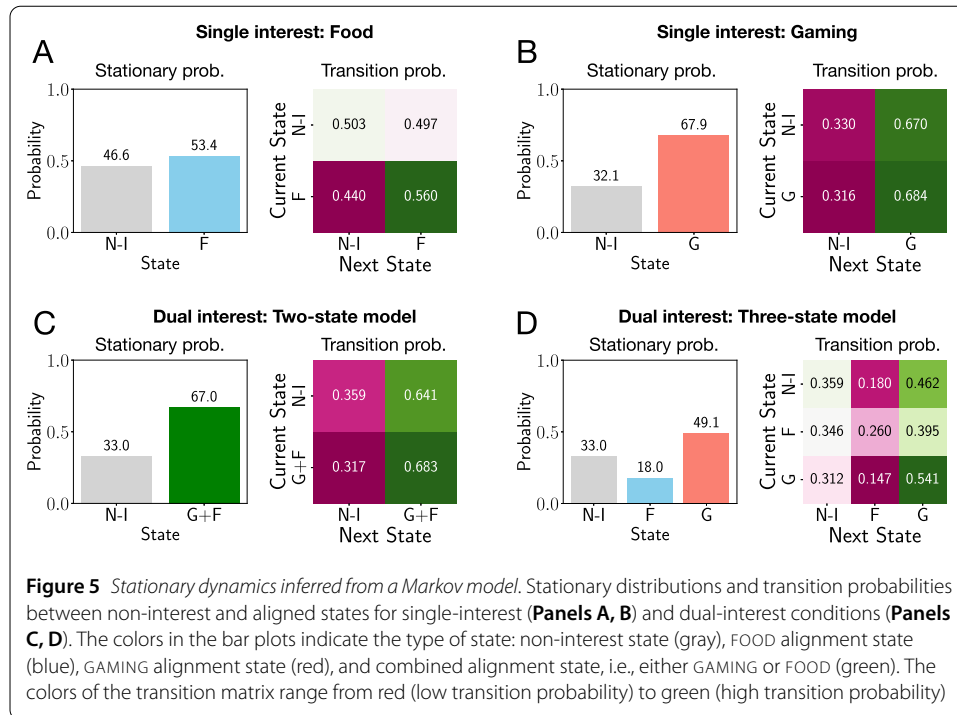
nounced in the GAMING condition, somewhat lower in the FOOD condition, and slightly more moderate for dual-interest bots (GAMING+FOOD). Notably, the counts $C(T)$ correspond to strikingly high ratios of interest-aligned content with 67.4% (GAMING), 52.3% (FOOD), and 67.2% (GAMING+FOOD).

Onset of content amplification We estimate the time at which TikTok's feed algorithm identifies the bots' interests as the point when it begins significantly increasing recommendations of interest-aligned content. To determine this, we apply an established change point detection method [25]. In Fig. 2, the onset times t_o are indicated by vertical black lines and star symbols.

The onset points are determined from the binary time-series $S_{\alpha,i}(t)$ and are typically located in regions where $r_{\alpha,i}(t)$ exhibits a sharp initial increase. Notably, most onset points occur within the first 200 videos viewed by the bots, though variations exist across experimental conditions, as depicted in Fig. 4. On average, the onset occurs after approximately 66 videos for single-interest GAMING bots, 140 videos for single-interest FOOD bots, and 93 videos for dual-interest GAMING+FOOD bots. Strikingly, this corresponds to (on average) only approximately 13 aligned videos before the onset of significant content amplification (at t_o).

After identifying the onset points of amplification, we can quantify how rapidly amplification unfolds, i.e., how quickly the system transitions to the high-amplification regime. We do so by calculating the slope of $r_{\alpha,i}(t_o)$ around the onset points t_o , denoted as $m_{\alpha,i} = [r_{\alpha,i}(t_o + d) - r_{\alpha,i}(t_o - d)]/2d$, where d represents the window size, i.e., the number of time steps (subsequent videos) used to compute the slope.

We find that the slopes $m_{\alpha,i}$ vary strongly across conditions $\alpha \in \{G, F, G + F\}$, as shown in the right panel of Fig. 4. The onset dynamics for GAMING content exhibit the steepest average slope (0.009), while FOOD content shows the lowest (0.005). For dual-interest bots, the average slope (0.008) falls between those of the two single-interest conditions. A larger (positive) slope indicates a faster transition. For example, the slope of 0.009 observed in the GAMING condition implies that the rate $r \in [0, 1]$ increases by 0.9 over the course of 100 encountered videos, transitioning from a low (e.g., $r = 0.05$) to a high alignment rate (i.e., $r = 0.95$). In contrast, the slope of 0.005 observed in the FOOD condition corresponds



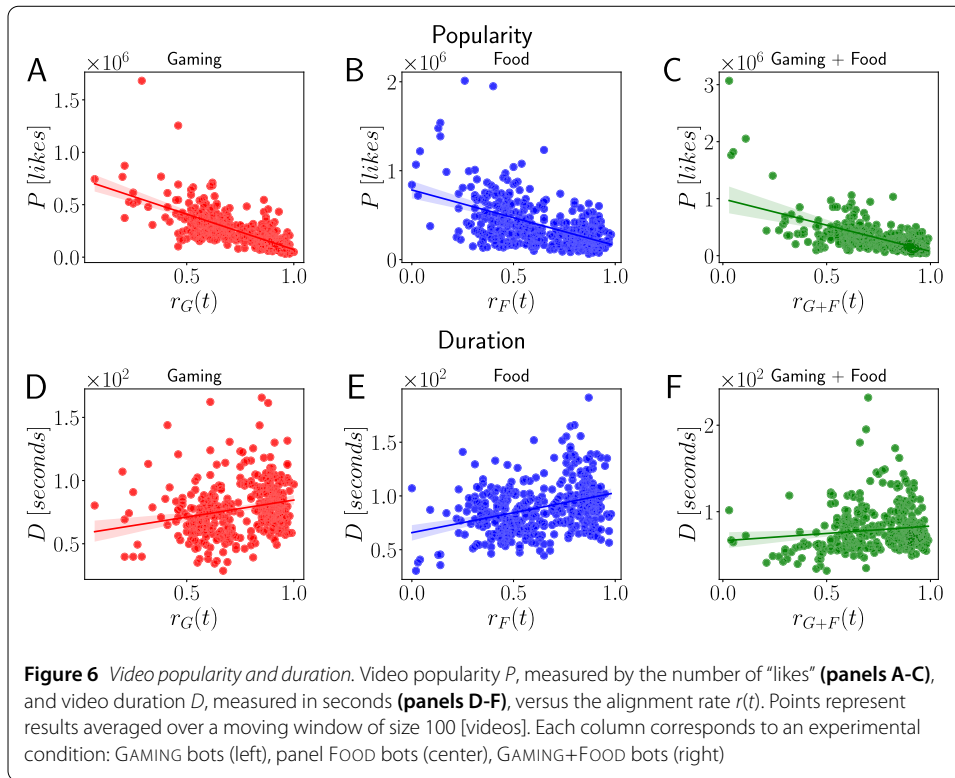
to a substantially slower transition, requiring nearly twice as many videos to achieve the same increase. See the SM for details on the fitting procedure to compute the slopes $m_{\alpha,i}$.

Post-onset dynamics To analyze the later-phase behavior of the recommendation dynamics, we apply a first-order Markov model [26]. Under the assumption of stationarity, we derive both the transition probabilities and the stationary probability distributions of the states under the model.

For single-interest bots, the system is modeled using two states: non-interest state (N-I, state 1), and aligned state (GAMING/FOOD) (state 2). For dual-interest bots, we examined two scenarios: (i) a *two-state* Markov model with non-interest state (state 1), and an aligned state, which aggregates GAMING- and FOOD content (state 2), and (ii) a *three-state* Markov model that distinguishes between non-interest state (state 1), and the two aligned states, GAMING (state 2), and FOOD (state 3), adding complexity but offering a more detailed representation of the content dynamics.

The results of the Markov model are shown in Fig. 5. For single-interest bots, the average transition probabilities exhibit a strong dependence on the topic of interest. In the FOOD condition, following the recommendation of a non-interest video, content aligned with the bots' interest appears 49.7% of the time. This probability increases to 56% after an interest-aligned recommendation, indicating a higher likelihood of FOOD content in subsequent video. Overall, the inferred transition probabilities of the Markov model give rise to an average stationary probability distribution of 46.6% for the non-interest state and 53.4% for the FOOD alignment state.

For single-interest GAMING bots, the Markov model reveals an even stronger amplification of interest-aligned content. Transitions from both states—non-interest state and alignment state—exhibit an elevated probability to be followed by interest-aligned content. Specifically, transitions from the non-interest state to the alignment state occur with



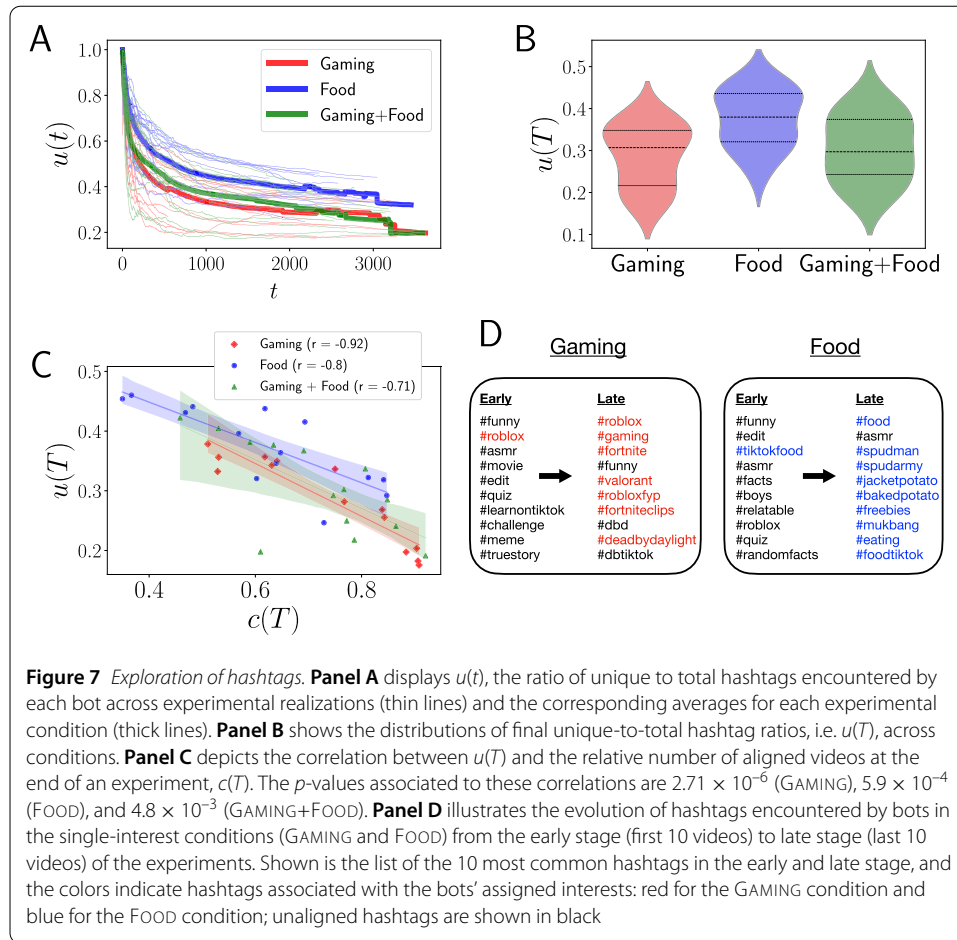
a probability of 67%, and interest-aligned content is followed by further interest-aligned content with an even higher probability of 68.4%. The average stationary distribution in the GAMING condition evaluates to 32.1% (non-interest state) and 67.9% (interest-aligned state).

The Markov model also reveals a strong amplification of interest-aligned content for the dual-interest bots. When aggregating the two interests (GAMING and FOOD) into a single state, we observe that interest-aligned content follows non-interest content in 64.1% of cases and follows interest-aligned content in 68.3% of cases, resulting in a stationary probability distribution of 33% for non-interest content and 67% for interest-aligned content.

The three-state Markov model allows for a more detailed analysis, giving insights into which topic might mostly drive the content amplification in the dual-interest condition. Indeed, the bottom-right panel of Fig. 5 suggests that GAMING content accounts for most of the amplification of interest-aligned content. Moreover, the transition probabilities reveal that transitioning to GAMING content is consistently the highest, regardless of the current state: 46.2% from non-interest content, 39.5% from FOOD content, and 54.1% from GAMING content. In contrast, FOOD content follows less frequently, with probabilities of 18% after non-interest content, 26% after FOOD content, and 14.7% after GAMING content. These transition dynamics result in a stationary distribution of 33% for non-interest content, 18% for FOOD content, and 49% for GAMING content, which is consistent with the two-state model.

Video duration and popularity We examined how video duration and popularity vary with the degree to which content aligns with the bots’ interests, as quantified by $r_{\alpha,i}(t)$.

Panels A–C in Fig. 6 present the results for video popularity P , which we approximate by the number of likes a video has received at the time it was encountered by the bot. Across



all experimental conditions, we observe a clear negative correlation between popularity and the rate of alignment: as r increases, i.e., content becomes more aligned with the bot's interests, its popularity decreases. Panels D-E summarize the findings for video duration D , measured in seconds. Here we observe the opposite trend. Specifically, across all three conditions, there is a strong *positive* correlation between $r_{\alpha,i}(t)$ and D : content that more closely matches a bot's interests tends to be longer than more general content.

Exploration of content To quantify exploration, we analyzed the hashtags associated with the videos encountered by the bots. Specifically, we examined the ratio of the number of unique hashtags $H_u(t)$, and the total number of hashtags $H_T(t)$ encountered by bots, i.e., $u(t) = H_u(t)/H_T(t)$. While the accumulation of unique hashtags (H_u) can be expected to follow patterns characteristic of real-world discovery processes, such as Heaps' or Zipf's law [27–30], the total number of encountered hashtags (H_T) will grow linearly at a rate proportional to the mean number of hashtags per video. Figure 7A shows the resulting growth dynamics of $u(t)$ across experimental realizations. In the figure, colors represent experimental conditions, thin lines correspond to individual experimental realizations, and thick lines indicate the averaged across realizations within each condition.

We find that the degree of exploration depends on the bots' interests: on average, the ratio $u(t)$ decays more rapidly for bots in the GAMING condition than for those in the FOOD condition, i.e., FOOD bots engage with a larger number of unique hashtags compared to

GAMING bots. The content exploration dynamics of dual-interest bots (GAMING+FOOD) generally fall between those of the two single-interest conditions. This contrast in content exploration is further highlighted in Fig. 7B, which depicts the distribution of final ratios of unique to total hashtag counts, $u(t = T)$, across experimental conditions.

We next examined the association between the amplification of content alignment and content exploration to investigate whether—and to what extent—a more aligned content feed reduces exploration at the hashtag level. Specifically, we examine the relationship between the normalized cumulative count of interest-aligned content, $c(T) = C(T)/T$, where $C(T)$ is the total number of aligned videos by the end of the experiment (T), and the normalized ratio of unique hashtags, $u(T)$ (as a measure for the degree of exploration). The results are shown in Fig. 7C. Across all three conditions, alignment and exploration exhibit strong negative correlations that are significant (p -values are reported in the caption of Fig. 7C). The correlation is strongest for bots interested in GAMING ($r = -0.92$) and weakest for those interested in GAMING+FOOD ($r = -0.71$), with FOOD bots falling in between ($r = -0.8$). Furthermore, examining the slopes of the linear fits for each condition provides additional insight. The GAMING condition exhibits the steepest negative slope, the FOOD condition shows the shallowest slope, and the GAMING+FOOD condition falls in between. Overall, steeper slopes reflect a greater decline in exploration as content becomes more aligned with a specific interest.

To better interpret the observed reduction in exploration with increasing content alignment, we examined the evolution of encountered hashtags over the course of the experiments. Focusing on the single-interest conditions (GAMING and FOOD), we aggregated all hashtags encountered by bots within each condition during the early stage (first 10 videos) and the late stage (last 10 videos) of each experimental run. The ten most frequently occurring hashtags from each stage are shown in Fig. 7D, with colored hashtags indicating topic-aligned ones. In both conditions, early-stage hashtags span a broad semantic space, including hashtags from more diverse topics. By contrast, late-stage hashtags show a clear convergence toward the bots' assigned interest, suggesting a narrowing of content exposure as alignment increases.

Importantly, both the amplification of aligned content and the reduction in exploration are influenced by social network factors (i.e., bots' following actions) as well as algorithmic factors (i.e., recommendations of content originating outside a bot's "social circle"). We conducted an additional analysis focusing on the proportion of videos encountered by each bot that originated from accounts it had previously followed. We define the time-dependent ratio of encountered videos from followed accounts as $f(t) = F(t)/t$, where t denotes the discrete time step (i.e., video index, $t \in \{1, 2, 3, \dots, T\}$) and $F(t)$ is the cumulative sum of videos a bot has encountered from accounts it had already followed up to time t . While $F(t)$ is a monotonically increasing function, $f(t)$ need not be. The final value $f(t = T)$ represents the overall fraction of videos recommended from followee accounts over the course of the entire experiment. On average, only approximately 10% of the videos engaged with by bots originated from followee accounts, suggesting that the vast majority of content exposure is driven by algorithmic recommendations rather than the bots' social network. The complete set of $f(t)$ curves for all experimental runs is provided in the SM (Fig. 4).

3 Discussion

In this work, we developed a sock-puppet methodology to investigate how TikTok's "For You" feed responds to bots' expressed interests. The behavior of the bots was conditioned on the content of videos: when a video was aligned with one or more interests of a bot, the bot performed a set of predefined actions to signal this interest. While these bot behaviors were too simplistic to fully mimic real human activity on TikTok, our sock-puppet audit was driven by a large language model, built on a bot-centric design, and capable of reacting to content in real time, which gives it great potential for future and more realistic studies of emergent phenomena in algorithmic systems.

Our results reveal that TikTok strongly amplifies content that is aligned with a bot's interests. Both single- and dual-interest bots experienced significant increases in the proportion of aligned content over time, with amplification factors varying across conditions. The onset of content amplification typically occurs within the first 200 videos, with bots interested in GAMING content exhibiting the fastest amplification and FOOD bots the slowest. After this initial phase, the recommendation dynamics exhibit distinct patterns that we uncovered and formalized using Markov models. Most importantly, we identified strongly increased transition probabilities toward interest-aligned content across all conditions. We also found that as content becomes more aligned, it becomes both less popular (lower like counts) and longer in duration, suggesting a potential trade-off between personalization and mainstream appeal. Finally, content exploration, as measured by the number of unique hashtags encountered by the bots, exhibits a strong negative correlation with content amplification: bots receiving more aligned content experience a lower diversity in hashtag exposure.

Our findings build on a growing body of research examining how algorithmic feeds shape and amplify content on social media platforms. Prior work has shown that recommendation systems can systematically reinforce certain types of content. For instance, observational analyses reveal that political content receives disproportionate amplification on Twitter [31], and that YouTube's recommendation system can steer users toward increasingly extreme material [32]. A related line of research employs controlled bots in sock-puppet audits to study algorithmic behavior on platforms such as YouTube, Twitter, and Google News [33–39]. These bots typically follow simple heuristics (e.g., random walks) or predetermined scripts, or they are "trained" through predefined search actions to mimic coarse user behaviors.

While such approaches are effective on more static platforms they are ill-suited for a fast-paced, video-first environment like TikTok, which is purely algorithmically-driven. Therefore, we developed a large-language-model-based sock-puppet system capable of classifying and responding to video content in real time as the bot scrolls through the feed. This introduces an important methodological advance over previous TikTok audits, where bots relied solely on matching predefined sets of hashtags in video descriptions [19]. Such hashtag-based systems can only detect interest-aligned content when it is explicitly marked (e.g., #food), whereas our LLM-based bots can recognize and react to semantically relevant content even in the absence of specific tags. This allows for a substantially more adaptive and ecologically valid audit of the platform's amplification dynamics.

In line with previous findings on Twitter [31], we find differences in the extent to which different topic get amplified on TikTok. Specifically, GAMING content undergoes greater amplification than FOOD content. Several factors may contribute to this phenomenon.

First, engagement-driven reinforcement may play a role, where higher user engagement (e.g., watch time, likes, shares) for certain content categories prompts the algorithm to prioritize these topics more aggressively. Second, differences in content supply dynamics may influence amplification. The prevalence of certain content types on the platform can shape the extent to which they are reinforced. Less frequently appearing content is less likely to be encountered by users, thereby reducing its potential for amplification. Finally, amplification may vary across content types, possibly due to variations in content classification accuracy or ranking mechanisms. GAMING content, for instance, may be amplified more because it is more narrowly defined and easier to classify compared to FOOD content. This raises broader questions about the amplification of other niche content, including potentially harmful material such as radical political videos, which could facilitate the formation of echo chambers and reinforce ideological polarization online [40–42].

Despite the strong amplification of interest-aligned content, exploration remains substantial. The stationary content distributions inferred through Markov models indicate that even in the strongest cases of amplification, roughly one-third of the content remains unaligned. This suggests that TikTok's feed algorithm indeed balances personalization and content diversity, prioritizing content alignment while preventing complete convergence. Additionally, the similar stationary probabilities observed between single- and dual-interest bots indicate that the algorithm does not necessarily amplify content more aggressively for bots with fewer interests. Importantly, we find strong negative correlations between the amount of aligned content a bot engages with during an experiment and the extent of its exploration within the hashtag space. These correlations show a potential downside of algorithmic amplification: as content becomes increasingly aligned, the exposure to diverse content on the level of individuals is strongly reduced [43].

Several limitations of this study should be acknowledged. While bots provide controlled insights into recommendation dynamics, they are nowhere near to perfectly replicate real user behavior, particularly in terms of more heterogeneous interactions with the platform. Each time a bot encountered aligned content, it consistently liked and rewatched the video and followed its creator, a pattern that does not reflect the variability of engagement typically observed in human users. Additionally, our bots did not analyze video content directly but instead inferred the topic of the video based on textual metadata, i.e., the video descriptions and hashtags. This reliance on textual cues introduces uncertainties in accurately identifying aligned content. Another challenge in quantifying content amplification in our experiments is the absence of a “neutral baseline,” i.e., bots *without* predefined interests and the content they would encounter over time. Due to TikTok's bot detection system, we were unable to collect such data, as interest-neutral bots were removed from the platform during the experiments. Most importantly, our study focused on only two content domains (GAMING and FOOD), which may not generalize to other topics.

Future research should extend this approach to politically sensitive or controversial content to examine whether similar amplification patterns emerge, and how those patterns differ across the globe, as we have focused on bots residing in the United States. Furthermore, designing more heterogeneous bots (including age, gender, and geographical location) that also express multiple, potentially competing interests, or bots that shift their interests over time, could help uncover how the algorithm modulates amplification when signals are less concentrated. While our experiments covered an extended period of time (roughly one week per bot), even longer-term studies could offer deeper insights

into whether amplification actually plateaus, continues to escalate, or oscillates over time. Finally, a more refined method for analyzing the actual video content in real time, e.g., through the use of multimodal models [44], would enable researchers to address more specific questions about algorithmic amplification on TikTok. For instance, the relatively large share of content classified as unaligned that persists even after amplification may in fact be semantically related to a bot's interests to a greater extent than our results suggest, as our classification relies on detection mechanisms based on textual cues alone.

This study provides a detailed empirical analysis of TikTok's recommendation dynamics, highlighting the rapid onset and sustained amplification of interest-aligned content. While strong content reinforcement occurs, TikTok's "For You" feed maintains a level of exploration that prevents complete collapse of content. The differential amplification across content types, the trade-off between personalization and content popularity, and the distinct pattern of recommendation revealed by Markov Models provide important insights into the mechanisms shaping algorithmic curation, and contribute to ongoing discussions about the societal role of algorithmic systems.

4 Methods

4.1 Sock-puppet audit design

To investigate TikTok's recommendation algorithm, we conducted a sock-puppet audit, deploying automated bots to interact with the platform's content feed under controlled conditions. Given TikTok's advanced bot detection mechanisms, we utilized Selenium as the primary tool for automation, supplemented with *Selenium Stealth* to minimize detectability and ensure that the bots' actions closely mimicked human-like behavior.

4.2 Experimental setup

All experiments were conducted within American time zones (9 AM to 2 AM EST) to align with realistic user activity patterns. To ensure exposure to U.S.-based content, we employed proxies to spoof the bots' locations to American servers.

Each bot was assigned one of three predefined interests: (i) GAMING, (ii) FOOD, or (iii) GAMING+FOOD (dual-interest). While initial tests explored a broader range of topics, these categories were selected due to their prevalence on TikTok, ensuring sufficient data collection for robust analysis.

Each experimental realization (i.e., each bot) spanned multiple days of TikTok browsing, with bots watching approximately 500 videos per day. These bot interaction rates were chosen to reflect empirical evidence that TikTok engagement is predominantly volume-driven rather than depth-driven: users typically scroll through large numbers of videos (often up to several hundred per day) while exhibiting limited engagement with content that does not align with their interests [17]. To minimize the risk of detection due to repetitive behavior, session lengths varied, and bots did not watch all 500 videos in a single sitting. Instead, they logged onto the platform for sessions of approximately 100 videos (± 10).

After each session, bots saved cookies, logged out, and remained inactive for a randomized period of 20–30 minutes before initiating the subsequent session. Each bot was associated with a distinct *Chrome profile*, allowing automatic re-authentication via stored credentials. This setup ensured seamless login, thereby circumventing TikTok's CAPTCHA verification.

Unless an earlier termination was required for other reasons, each bot was run for approximately one week (i.e., 7 days), resulting in over 3000 videos watched in most cases.

4.3 Bot behavior and interaction

Bots were created manually, each assigned a unique name and email address. To generate names for TikTok accounts, we randomly selected from a dataset provided by the Python name API. All bots were registered with a male gender designation, as the influence of gender on content amplification was not the focus of this study. A complete list of experimental realizations and associated statistics is provided in the SM.

During each session, bots logged into TikTok, viewed video streams, and engaged in user-like interactions based on the encountered content. Engagement decisions were guided by a language model (*GPT-3.5 Turbo*), which evaluated video descriptions and hashtags to determine their relevance to the bot's assigned interests. To assess classification reliability, we evaluated GPT-3.5 against a blind hand-coded sample of 500 randomly selected videos. The model achieved an accuracy of approximately 92% in distinguishing interest-aligned from non-aligned content. To maintain consistency with an English-speaking user profile, videos in languages other than English were skipped. The specific prompts provided to GPT-3.5 Turbo for content evaluation and language identification are detailed in the SM.

If a video was deemed relevant, the bot engaged with it by performing the following three actions: (i) liking the video, (ii) rewatching it a randomly determined number of times between 1 and 3, and (iii) following the content creator. Conversely, for non-relevant videos, the bot minimized engagement by watching only 5–10% of the total video duration before scrolling to the next video.

4.4 Empirical measures

Quantifying aligned content over time For visualization purposes (Fig. 2), we compute the rate of interest-aligned content over time. For each experimental realization i and condition $\alpha \in \{G, F, G + F\}$, the data are represented as a binary discrete-time series $S_{\alpha,i}(t) \in \{0, 1\}$, where $t \in \{1, \dots, T\}$ indexes the videos encountered by the bot from the first ($t = 1$) to the last ($t = T$). The value $S_{\alpha,i}(t) = 1$ indicates that the t -th video is aligned with the bot's assigned interest in condition α , and $S_{\alpha,i}(t) = 0$ otherwise.

The rate of interest-aligned videos over time, $r_{\alpha,i}(t)$, is then defined as a time-moving average of $S_{\alpha,i}(t)$. Formally, we compute the discrete convolution

$$r_{\alpha,i}(t) = \sum_{k=-\lfloor w/2 \rfloor}^{\lfloor w/2 \rfloor} \frac{1}{w} S_{\alpha,i}(t-k),$$

where w denotes the width of the averaging window. This smoothed rate $r_{\alpha,i}(t)$ captures the local fraction of aligned videos around time t and serves as our primary measure of interest-specific amplification.

In addition to the moving-average rate $r_{\alpha,i}(t)$, we track the cumulative number of aligned videos encountered by each bot. For a given condition α and experimental realization i , the cumulative count is defined as

$$C_{\alpha,i}(t) = \sum_{\tau=1}^t S_{\alpha,i}(\tau),$$

which records how many of the first t recommended videos match the bot's assigned interest. The total number of aligned videos seen during an entire experimental run is thus

$C_{\alpha,i}(T)$. In Fig. 2, we display $C_{\alpha,i}(t)$ as a dashed red curve on a logarithmic scale on the right y -axis, complementing the smoothed rate $r_{\alpha,i}(t)$ shown on the left y -axis. For the dual-interest condition (GAMING+FOOD), we further plot the cumulative counts and rates for GAMING and FOOD separately, alongside their combined trajectory, to illustrate how the recommendation system responds to multiple expressed interests.

Baselines To quantify amplification, we compare the observed rate of aligned content to two empirical baselines.

Early-phase baseline. For each experimental run i and topic α , we define

$$b_1^{\alpha,i} = \frac{C_{\alpha,i}(t_0)}{t_0},$$

where t_0 is the estimated change point at which amplification begins and $C_{\alpha,i}(t_0)$ is the cumulative number of aligned videos up to that point (see above). This baseline captures exposure to aligned content before the system infers and reinforces the bot's interest.

Cross-condition baseline. For each topic $\alpha \in \{\text{GAMING}, \text{FOOD}\}$, we define a second baseline b_2^α that measures how often—on average— α -content appears when it is *not* the bot's assigned interest. For example, the baseline for GAMING is computed from the frequency of GAMING videos encountered by FOOD bots, and vice versa. For the dual-interest condition, we use the mean of the two single-interest baselines. This baseline approximates the background prevalence of a topic in the feed in the absence of explicit interest signaling.

These two baselines together approximate the non-interest regime of the platform: the early-phase baseline characterizes pre-amplification exposure for each bot, while the cross-condition baseline characterizes topic prevalence when it is not the focal interest.

4.5 Data collection and storage

The study was conducted over three months [June ('2024-06-04 15:00:53')–August ('2024-08-18 18:52:55')], during which seven test cycles were completed, deploying a total of 42 bots. Each test lasted approximately 6–7 days, with each bot watching an average of 3000 videos per test cycle.

In total, the bots processed 128,148 videos, of which 66,672 were unique. For analysis, we excluded videos lacking descriptions or hashtags, resulting in a final dataset of 112,610 videos, including 58,213 unique entries. The cleaned dataset contained a total of 817,200 hashtags, of which 63,694 were unique. Additionally, the dataset included 27,507 distinct content creators, with an average video duration of 79 seconds. On average, bots watched approximately 64% of each video's total length.

All collected data, including video-specific details such as duration, description, and hashtags, along with additional metadata such as the content creator's name and the number of likes, were stored in a *MongoDB Atlas* database for subsequent analysis.

Abbreviations

SM, Supplementary Material.

Supplementary information

Supplementary information accompanies this paper at <https://doi.org/10.1140/epjds/s13688-026-00629-2>.

Additional file 1. (PDF 3.7 MB)

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Author contributions

F.B. and A.C. designed the research, analyzed the data, and wrote the main manuscript. N.A. developed the sock-puppet methodology. All authors reviewed the manuscript.

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Data availability

Anonymized data that support the findings of this study will be made publicly available in an open-access repository upon publication. A link to the repository will be provided in the final version of the manuscript.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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